

Evaluating the Fed

How good are the forecasts of policy makers?

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This was a four-author project (M. Chittineni, P. Eremeev, L. Xie, C. Zhang). My contributions are Section 3 (Data and Methodology), Section 4 (Results: Absolute Forecast Accuracy), and Section 5 (Benchmarked Performance).

1 History of Forecasts

To better understand the accuracy of forecasts made by policymakers, it is essential to review their brief history. We will focus primarily on the Federal Reserve (Fed), while also providing a concise introduction to the history of projections beyond the Fed.

Until 1907, when a financial crisis caused by runs on trust companies was coordinated privately by J.P. Morgan, Congress did not realize that a systemic institutional solution was indispensable. This realization led to the establishment of the National Monetary Commission, whose recommendations eventually facilitated the creation of the Federal Reserve in 1913. At its inception, the Fed was not independent. Both the Secretary of the Treasury and the Comptroller of the Currency were members of the Board; as executive officials, they reported directly to the President. The government maintained institutional influence over the Fed, allowing the President to exert control through the Treasury Secretary and the Comptroller, who were keen to retain their authority over the institution.

In 1933, Franklin D. Roosevelt was elected as the 32nd President of the United States. In 1934, he appointed Marriner Eccles as Chairman of the Federal Reserve Board. Eccles proposed formally placing the Fed under presidential control, stating he would resign upon the inauguration of a new president. However, Congress rejected this proposal. Members of Congress observed a recurring pattern: politicians naturally lean toward loose monetary policy—lowering interest rates and expanding credit can provide short-term economic stimulus and aid elections, but at the cost of long-term inflation, the political burden of which often falls on the subsequent administration. Consequently, in 1935, during the Roosevelt administration, Congress reorganized the Fed, removing the Secretary of the Treasury and

the Comptroller of the Currency from the Board. During the 1935 hearings, nearly 60 witnesses almost unanimously opposed presidential control of monetary policy; even the then-Treasury Secretary Morgenthau expressed support for an independent agency. Progress toward Fed independence nearly regressed during World War II, as the Treasury demanded the Fed maintain low interest rates to finance the war.

In 1945, Harry S. Truman succeeded as the 33rd President. Subsequently, in 1946, the Council of Economic Advisers (CEA) was established within the Executive Office of the President under the Employment Act. This was the U.S. government's first institutionalized economic forecasting body, tasked with providing the President with objective analysis and advice on economic policy and preparing the annual Economic Report of the President. These reports contained limited economic outlooks covering forecasts for GDP growth, unemployment, and inflation.

Following the outbreak of the Korean War, inflationary pressures rose, and the Fed sought to end the interest rate pegging policy. In 1951, President Truman falsely informed the media that the FOMC had agreed to support the Treasury's pegging policy; Eccles immediately countered by leaking meeting minutes to the New York Times. Ultimately, both parties reached the "Treasury-Fed Accord," granting the Fed the power to determine interest rates independently. This year marked the beginning of true Fed independence.

Prior to the creation of the "Greenbook" in 1964, forecasts within the Fed were loose and lacked a systematic structure. Independent institutions in a democratic system naturally face questions of legitimacy. An independent body determining monetary policy is a cause for concern. The Fed's transparency has often been a reactive response to legislative pressure rather than a proactive choice. As stated in a 2014 speech by the Cleveland Fed: "Independence in setting monetary policy is worth preserving because it yields more effective policy. But accountability must go hand-in-hand with independence. A central bank cannot expect to remain independent from the political process unless it is transparent about the basis for its policy decisions." Furthermore, transparency serves as a protection for the Fed, ensuring the public can verify that the institution maintains its independence. Therefore, the accuracy of forecasts is not merely a technical issue, but a "performance indicator" for the independent institution to gain political legitimacy. If forecasts deviate from reality over the long term, the foundation of public trust in the Fed's independent exercise of power will be shaken.

Subsequently, the Fed's transparency increased incrementally. In 1975, as the U.S. economy bottomed out after the severe recession following the 1973 oil crisis, congressional anxiety regarding the Fed's actions reached a peak. That year, Congress passed HCR-133, a Concurrent Resolution requiring the Fed to report monetary growth targets to Congress regularly.

In 1976, the OMB (Office of Management and Budget) included ten-year economic forecasts in the President's annual budget; these forecasts were embedded directly into the budget proposal, which is the core document of fiscal policy. The

OMB's forecasts and fiscal policy decisions are nearly integrated. In 1978, the Humphrey-Hawkins Act was formally enacted, clarifying that the Fed must not only report monetary targets but also submit comprehensive economic forecasts and testify to Congress regarding employment and inflation goals. This directly catalyzed the first release of semiannual economic projections in 1979. In 1983, the first publication of the Beige Book occurred, summarizing information about economic conditions received from the Federal Reserve System's business contacts.

In the 1990s, with increasing market complexity, the Fed realized that information asymmetry itself could trigger financial instability. Before 1994, the Fed did not issue an immediate statement after adjusting interest rates; market participants had to deduce policy intentions by observing open market operations. This created severe information asymmetry, allowing certain institutions with professional analytical capabilities to profit ahead of the public and amplifying market volatility. In 1994, the decision was made to release a post-meeting statement when policy actions were taken; in 2000, the practice began of issuing a statement after every FOMC meeting, including an assessment of the balance of risks to the Committee's objectives.

To enhance transparency and individual accountability, the FOMC roll call vote was added to the post-meeting statement in 2002. In 2005, the release of FOMC minutes was accelerated from a delay of six or seven weeks to just three weeks. The early release of minutes addressed the issue of "information timeliness decay." Under the old system, by the time minutes were released, the next policy meeting was often imminent, rendering the minutes historical documents. Shortening the delay to three weeks transformed the minutes into key clues for the market to understand internal disagreements and future policy paths. Research (such as "The Financial Market Effect of FOMC Minutes" by the New York Fed) indicates that the release of minutes induces significant asset price volatility, particularly in short-term Treasury yields, suggesting that the forecast information in the minutes was not fully covered by previous statements.

The 2008 financial crisis acted as a catalyst. Following the crisis, the Fed introduced unconventional tools like QE, and "Forward Guidance" transitioned from vague ("at least through 2015") to conditional, and then to numerical thresholds (6.5% unemployment / 2.5% inflation). This marked a critical shift in how projections evolved from "forecasts" into "policy tools" themselves, greatly increasing the weight of communication in economic stimulation. However, this also introduced a severe evaluation challenge: if forecasts ultimately prove inaccurate, the Fed faces a significant test of its credibility (as noted in "The Effectiveness of 'Forward Guidance' during the Great Recession" by the Philadelphia Fed).

In 2011, the Federal Open Market Committee (FOMC) began publishing the Summary of Economic Projections (SEP), which systematically reports participants' projections for GDP growth, the unemployment rate, headline and core PCE inflation, and the appropriate path of the federal funds rate (Federal Open Mar-

ket Committee, Summary of Economic Projections, January 2012). In 2012, the FOMC added individual interest rate projections—commonly referred to as the “dot plot”—to the SEP, and in the same year adopted the Statement on Longer-Run Goals and Monetary Policy Strategy, which formally established a 2 percent longer-run inflation objective (Federal Open Market Committee, Statement on Longer-Run Goals and Monetary Policy Strategy, January 24, 2012). This statement provided an explicit nominal anchor for the interpretation of economic projections. In 2020, the FOMC revised the strategy statement to indicate that inflation would aim to average 2 percent over time, noting that following periods in which inflation has run persistently below target, appropriate monetary policy may allow inflation to run moderately above 2 percent for some time (Federal Open Market Committee, Statement on Longer-Run Goals and Monetary Policy Strategy, August 27, 2020).

2 Why should we care about forecasts?

The Federal Reserve’s ability to fulfill its dual mandate, maximum employment and stable prices, depends not only on the policy actions it takes but on the expectations of other economic agents who observe it. If market participants believe the Fed to be a poor forecaster, the signaling value of its guidance diminishes; if they believe it to be a reliable one, expectations alone can shift behavior before any policy instrument is actually felt on economic activity.

This asymmetry between the speed of policy transmission and the speed of belief updating gives a sufficient condition for us to care about forecast accuracy. Conventional monetary policy operates with lags, changes to the federal funds rate or effects of LSAP propagate through financial markets, investment decisions, and aggregate demand over many months. Expectations, by contrast, update instantaneously. Households and firms making consumption and investment decisions condition those decisions on their beliefs about the future path of inflation and output growth. The basic consumer optimization model explicitly defines the choice of consumption today as a function of future income. A credible Fed anchors these beliefs. The Fed therefore stands to gain from being perceived as an accurate forecaster nearly as much as it stands to gain from being one.

3 Data and Methodology

3.1 Forecast Data

The forecast data consist of digitized Greenbook and Tealbook projections produced by Federal Reserve Board staff prior to each FOMC meeting, compiled by the Philadelphia Federal Reserve. We chose to focus on projections for three variables: the unemployment rate (UNEMP), CPI inflation expressed as a quarterly annualized growth rate (gPCPI), and real GDP growth similarly expressed (gRGDP). These

were chosen because their union broadly covers both sides of the dual mandate, and because variability of economic indicators is plausibly proportional to the magnitude of economic activity.

We evaluate forecasts at two horizons. The nowcast, denoted F_0 , is the staff's estimate of the current quarter, a quarter for which the official data has not yet been released. The four-quarter-ahead forecast, denoted F_4 , represents the staff's projection one year into the future. The digitized dataset supports forecasts up to F_9 (approximately 2.25 years), but sample size drops sharply beyond F_4 . For unemployment, available observations fall from 407 at F_4 to 96 at F_9 ; for CPI inflation, from 325 to 32; and for real GDP growth, from 407 to 96. These reductions are of approximately 76%, 90%, and 76% respectively. As such, our selection of F_0 and F_4 was driven not by methodological considerations but by the constraints of the dataset we chose to work with.

3.2 Realized Data

Realized values are drawn from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis. For unemployment we use the civilian unemployment rate (UNRATE); for the price level, the Consumer Price Index for All Urban Consumers (CPIAUCSL); and for output growth, the quarterly annualized percent change in real GDP (series A191RL1Q225SBEA).

A measurement issue arises for inflation. The Greenbook reports CPI inflation as a quarterly annualized growth rate, while CPIAUCSL is a price index level. We need to construct a comparable inflation series. Thus we compute,

$$\pi_t = \left[\left(\frac{\text{CPI}_t}{\text{CPI}_{t-1}} \right)^4 - 1 \right] \times 100$$

We have CPI_t denoting the index value in quarter t , the ratio $\text{CPI}_t/\text{CPI}_{t-1}$ gives the gross quarterly growth rate, the exponent of 4 annualizes this quarterly rate, subtracting 1 converts from gross to net growth, and multiplication by 100 converts into percentage points.

A second issue concerns data vintage. FRED revises historical data over time; this analysis uses the most recent vintage of realized values, not the real-time data available at the time each forecast was made which would have to be implemented through ALFRED. As a consequence, our results are prone to overstate forecast error for early periods, since those data points have undergone more cumulative revision. Likewise, any benchmark forecasts like naive and AR(1) are also estimated on the same final-vintage FRED, meaning they benefit from revised history that was not available in real time. This slightly advantages the benchmarks relative to the Greenbook, making any measured Greenbook performance a conservative estimate. The logic was that it improves the data for regression processes, and corrects for improvements that arose from data quality, instead of forecasting techniques.

3.3 Error Metrics

Let y_{t+h} denote the realized value of a variable h quarters after the forecast origin t , and let $\hat{y}_{t+h|t}$ denote the forecast made at time t for that realization. The forecast error is

$$e_t = y_{t+h} - \hat{y}_{t+h|t}$$

We summarize forecast performance using two standard loss functions. The mean squared error over N forecast origins is

$$\text{MSE} = \frac{1}{N} \sum_{t=1}^N (y_{t+h} - \hat{y}_{t+h|t})^2$$

and the mean absolute error is

$$\text{MAE} = \frac{1}{N} \sum_{t=1}^N |y_{t+h} - \hat{y}_{t+h|t}|$$

Both treat overestimation and underestimation symmetrically, meaning error is penalized uniformly regardless of sign. This is a reasonable source of potential deviation. Likewise, MSE penalizes large errors disproportionately due to the squared term and is therefore more sensitive to outliers, which is why we partition some of our basic analysis into including/excluding COVID. MAE treats all errors linearly and thus when it comes to outliers provides a better measure of central tendency in forecasts.

To track how accuracy evolves over the sample, we compute a rolling moving average of absolute errors centered at each forecast date t :

$$\text{MAE}_t^{(n)} = \frac{1}{n} \sum_{i=t-\lfloor n/2 \rfloor}^{t+\lceil n/2 \rceil-1} |y_{i+h} - \hat{y}_{i+h|i}|$$

where n is the window width (here, the number of forecasts spanning five years or 40 data points), and the floor $\lfloor \cdot \rfloor$ and ceiling $\lceil \cdot \rceil$ functions handle the case where n is odd, yielding an approximately centered window. The result is a smoothed time series of mean error that is easier to interpret visually than the raw absolute errors.

3.4 Linear Trend

As part of our visualization and as a component of an AR(1) process, we utilize Ordinary Least Squares. We calculate the estimated slope as,

$$\hat{\beta} = \frac{\sum_{i=1}^N \left(t_i - \frac{1}{N} \sum_{j=1}^N t_j \right) \left(|y_{i+h} - \hat{y}_{i+h|i}| - \frac{1}{N} \sum_{k=1}^N |y_{k+h} - \hat{y}_{k+h|k}| \right)}{\sum_{i=1}^N \left(t_i - \frac{1}{N} \sum_{j=1}^N t_j \right)^2}$$

The components are worth defining and are as follows. t_i is the date of the i -th Greenbook forecast. The term $\frac{1}{N} \sum_{j=1}^N t_j$ is the mean forecast date across the entire sample, so $(t_i - \frac{1}{N} \sum_{j=1}^N t_j)$ is the forecast date minus the mean of forecast dates, or how far each forecast date lies from the sample midpoint. In the numerator, $|y_{i+h} - \hat{y}_{i+h|i}|$ is the absolute forecast error for the i -th observation, and $\frac{1}{N} \sum_{k=1}^N |y_{k+h} - \hat{y}_{k+h|k}|$ is the mean absolute error across the entire sample, so their difference is the deviation from the mean of the absolute error. The numerator is therefore the sample covariance between forecast dates and absolute errors; the denominator is the sample variance of the forecast dates. In our case, we hope for a negative $\hat{\beta}$, as that indicates that absolute errors have tended to decrease over time, meaning accuracy is improving.

For visualization this model is sufficient, but note that for multistep forecasts where we iterate over the same data multiple times (such as the F_4 horizon), forecast errors span overlapping time periods and are therefore correlated, following approximately an $MA(h-1)$ process. The OLS slope $\hat{\beta}$ remains unbiased; it should still correctly identify the direction and magnitude of the visual trend. However, the point is that OLS standard errors are inconsistent, meaning we are unable to perform any hypothesis testing on $\hat{\beta}$.

3.5 Benchmark Models

We compare the Greenbook against two mechanical benchmarks to assess the value added by the Fed's institutional forecasting apparatus.

Naive (random walk). The naive forecast sets

$$\hat{y}_{t+h|t} = y_t$$

That is, the best forecast of any future value is simply the most recent observation. This is a nonparametric estimate and represents a no-information lower bound on forecast performance. If the Fed is unable to beat this estimate, the value they add to the process would be questionable. With that being said, for smoother variables such as unemployment, the naive forecast can perform reasonably well at short forecast horizons.

AR(1). We model the process as a first-order autoregression:

$$y_t = c + \phi y_{t-1} + \varepsilon_t$$

where c is a constant (capturing the unconditional mean of the process when $|\phi| < 1$), ϕ is the autoregressive coefficient and ε_t is a white noise error term. This is provided for completeness, as ε_t is an unobservable "structural" noise.

The parameters are estimated by OLS using an expanding window. For each forecast origin t_i , we fit c and ϕ on all data available up to but not including the forecast date:

$$[\hat{c}, \hat{\phi}] = \arg \min_{c, \phi} \sum_{s=1}^{t_i-1} (y_s - c - \phi y_{s-1})^2$$

This means that as t_i increases through iterating newer sample data, the sample size grows, and so the model becomes "better trained" as it benefits from an increasing amount of historical data.

The h -step-ahead forecast is generated by iterating the estimated AR(1) forward:

$$\hat{y}_{t_i+h|t_i}^{(s)} = \hat{c} + \hat{\phi} \hat{y}_{t_i+h|t_i}^{(s-1)}, \quad s = 1, \dots, h$$

Since the first data point has nothing to regress on, we use the initial condition $\hat{y}_{t_i|t_i}^{(0)} = y_{t_i}$. At each step s , the model takes the previous iteration's forecast and applies the estimated autoregressive relationship to project one additional quarter forward. After h such iterations, the result is the AR(1) forecast for $t_i + h$.

For both benchmarks, we compute the ratio of Greenbook error to benchmark error:

$$\text{Ratio} = \frac{\text{MAE}_{GB}}{\text{MAE}_{BM}}$$

where the numerator and denominator are the MAE (or analogously the MSE) of the Greenbook and benchmark respectively. This metric can likely be improved on, but the benefit is interpretation; below 1 indicates the Greenbook outperforms the model; the further below 1, the greater the advantage we assign to the Fed's models and likely nonpublic data.

4 Results: Absolute Forecast Accuracy

4.1 Unemployment

The unemployment nowcast gives MSE of 0.03 and MAE of 0.13 pp (Figure 1), which makes sense as unemployment leverages within-quarter data releases like unemployment claims. The signed errors (Figure 2) are approximately mean-zero but spike around the 1974–75, 1980–82, and 2008 recessions, where the Fed underestimated the speed of labor market deterioration. The trend slope is -0.0011 pp/year (Figure 3), which seems negligible. At F_4 , MSE rises to 1.29 and MAE to 0.66 pp. The trend slope of $+0.0062$ pp/year is the only positive value in the sample (Figure 4), seemingly driven by COVID-era errors. Excluding COVID data reverses the sign to -0.0034 (Figure 5).

4.2 CPI Inflation

The CPI nowcast yields MSE of 0.83 and MAE of 0.62 pp. The rolling MAE (Figure 6) tracks macroeconomic eras: elevated during the disinflation period, minimal through the “Great Moderation”, and rising again mid-2000s. The trend slope of -0.0077 pp/year is the steepest in the sample, although we are not sure whether it reflects inflation itself becoming more stable or improvements in forecasting. At F_4 , MSE rises to 4.05 and MAE to 1.46 pp (trend: -0.0020 pp/year; -0.0064 excluding COVID).

4.3 Real GDP Growth

GDP growth is the most volatile variable. The nowcast MSE of 5.76 exceeds unemployment’s by orders of magnitude. At F_4 (Figure 7), MSE reaches 19.45 and MAE 2.59 pp. The Greenbook F_4 forecast is visibly smoother than the real data. The error series (Figure 8) seems to be strongly negatively skewed, with recessions producing the largest absolute misses. The trend slope is -0.0043 pp/year (Figure 9), but as with CPI, we are unsure whether the decline reflects forecasting improvements or simply the reduction in output volatility characteristic of modern eras.

4.4 Summary

Table 1: Greenbook Absolute Forecast Errors

Variable	Horizon	N	MSE	MAE
Unemployment	F_0 (Nowcast)	482	0.03	0.13 pp
Unemployment	F_4 (1-Yr Ahead)	407	1.29	0.66 pp
CPI Inflation	F_0 (Nowcast)	325	0.83	0.62 pp
CPI Inflation	F_4 (1-Yr Ahead)	325	4.05	1.46 pp
Real GDP Growth	F_0 (Nowcast)	482	5.76	1.85 pp
Real GDP Growth	F_4 (1-Yr Ahead)	407	19.45	2.59 pp

Table 2: Trend of Absolute Forecast Errors (Percentage Points / Year)

Variable	Horizon	Trend Slope	Excl. COVID Slope
Unemployment	F_0 (Nowcast)	-0.0011	N/A
Unemployment	F_4 (1-Yr Ahead)	$+0.0062$	-0.0034
CPI Inflation	F_0 (Nowcast)	-0.0077	N/A
CPI Inflation	F_4 (1-Yr Ahead)	-0.0020	-0.0064
Real GDP Growth	F_0 (Nowcast)	-0.0291	N/A
Real GDP Growth	F_4 (1-Yr Ahead)	-0.0043	-0.0450

The ranking across variables (unemployment, CPI, GDP growth) is at least consistent across both horizons and both error valuation functions. All trend slopes are small relative to the level of errors. In general, the declining error trends for some variables are plausibly related to reductions in macroeconomic volatility, not necessarily improvements in forecasting methodology.

5 Benchmarked Performance

The absolute errors in Section 3 are essentially meaningless without something to compare to. We benchmark the Greenbook against the naive random walk and the AR(1) model defined in the earlier section.

5.1 Naive Benchmark

The Fed outperforms the naive forecast across all six variable-horizon pairs. The MAE ratios range from 0.38 (CPI F_0) to 0.74 (GDP F_4); MSE ratios from 0.14 to 0.65 (Figures 10, 11). Two patterns seem notable. First, the Fed’s advantage is largest for inflation nowcasts and smallest for GDP at longer horizons, where the measured series is most volatile. Second, the MSE ratios are lower than the MAE ratios, plausibly suggesting that the Fed’s relative advantage is concentrated in avoiding large errors rather than improving median performance, since MSE, as discussed, punishes outliers more heavily. This is consistent with the Fed’s reliance on judgment calls that could help avoid catastrophic misses, even when the typical forecast is not that much better than a naive extrapolation. The time-series overlays (Figures 12, 13) reinforce this. For unemployment F_0 , the Greenbook and naive rolling MAEs both remain low, but the naive error spikes sharply around recessionary periods while the Greenbook remains stable. For GDP F_4 , the two series track each other more closely; the Fed’s edge narrows as the horizon extends and the underlying long-run trends are masked in random noise.

5.2 AR(1) Benchmark

Against the AR(1), margins narrow considerably (Figures 14, 15). MAE ratios range from 0.41 (CPI F_0) to 0.98 (GDP F_4). The GDP F_4 result is surprising: the Greenbook’s MAE of 2.59 pp is practically identical to the AR(1)’s 2.64 pp (Figure 16). One year ahead, the Fed’s institutional advantages add almost nothing over a two-parameter autoregression for GDP growth. This is not because the AR(1) is a good forecaster in absolute terms; both produce large errors. For unemployment and CPI, the Fed’s advantage remains meaningful, particularly at the nowcast horizon. The CPI F_0 ratio of 0.41 against the AR(1) (Figure 17) suggests the Fed’s access to current information metrics adds forecasting power for current-quarter inflation estimation. At F_4 , the CPI ratio rises to 0.91, indicating that the Fed’s advantage dissipates as the horizon grows.

5.3 Interpretation

Table 3: Greenbook / Benchmark MAE Ratios

Variable–Horizon	vs. Naive	vs. AR(1)
Unemployment F_0	0.56	0.55
Unemployment F_4	0.64	0.67
CPI Inflation F_0	0.38	0.41
CPI Inflation F_4	0.77	0.91
Real GDP Growth F_0	0.61	0.78
Real GDP Growth F_4	0.74	0.98

Table 3 summarizes. The Fed consistently beats both benchmarks, but the margin is horizon and variable dependent. The Fed’s value-added is largest where it has the most informational advantage (short-horizon, inflation) and smallest where the forecasting problem is hardest (long-horizon, GDP growth). The gap between the naive and AR(1) ratios is itself informative: for unemployment F_0 , both ratios are nearly identical (0.56 vs. 0.55), meaning the AR(1) adds little over the naive for a persistent, slow-moving variable. For GDP F_4 , the AR(1) closes much of the gap (0.74 to 0.98), implying that some of the Fed’s edge over the naive may be attributable to the Fed itself relying on something like the simple autoregressive structure, rather than to institutional advantage.

6 Market Evaluation of Inflation forecasts

A meaningful external comparison is to evaluate the Greenbook’s CPI F4 forecast against the Cleveland Fed’s 1-year expected inflation series (EXPINF1YR), drawn from FRED alongside the CPIAUCSL series used throughout this paper. Unlike long-maturity breakeven, it estimates near-term expected inflation directly by combining Treasury yields, inflation swaps, and survey measures.

This comparison is motivated by the narrowing Greenbook advantage at F4: the CPI MAE ratio rises to 0.77 against the naive benchmark and 0.91 against the AR(1). One-year-ahead inflation is therefore the natural place to ask whether the Fed adds value beyond what is already priced by markets. Over the overlap sample (1982Q1–2019Q4, $N=152$), the market series produces an MAE of 1.41 pp and MSE of 4.03, against the Greenbook’s 1.46 pp and 4.05. The resulting Greenbook-to-market ratio is 1.04 on MAE and 1.01 on MSE. Unlike every other comparison in this paper, the Greenbook does not outperform and the market is marginally more accurate on both measures.

The sub-period breakdown reveals where the divergence lies. During the disinflation era, the market MAE is 1.69 pp with a systematic upward bias of 0.63 pp. This is the period where the Greenbook’s structural judgment most plausibly adds

value. In both the Great Moderation and post-GFC periods, market MAE falls to 1.32 pp with near-zero bias, reflecting efficient price discovery in stable inflation environments.

Table 4: Greenbook vs. Market — CPI F_4 (1982Q1–2019Q4, N = 152)

	MAE (pp)	MSE	MAE Ratio	MSE Ratio
Greenbook	1.46	4.05	—	—
Market (EXPINF1YR)	1.41	4.03	1.04	1.01

Table 5: Market CPI F_4 MAE by Sub-Period

Period	N	MAE (pp)	Bias (pp)
Disinflation (1982–1990)	36	1.69	−0.63
Great Moderation (1991–2007)	68	1.32	−0.16
Post-GFC (2008–2019)	48	1.32	+0.03
Full overlap (1982–2019)	152	1.41	−0.21

Two caveats apply. First, EXPINF1YR is not a pure CPI forecast- it embeds risk premia and liquidity effects. So, this comparison is best read as Greenbook versus market-implied expectations rather than a like-for-like test. Second, the series has been revised over time, introducing vintage issues analogous to those in Section 3.2; this slightly disadvantages the Greenbook, making any underperformance a conservative estimate. No comparable exercise is feasible for unemployment or GDP, where no market instrument directly reveals expectations in the way inflation-linked securities do.

Lastly, no comparable market instrument directly prices expected future unemployment or GDP growth the way inflation-linked securities reveal inflation compensation. Survey-based measures like the SPF do exist for both variables, but benchmarking against surveys rather than market prices would introduce a different and harder-to-justify methodological equivalence, and is beyond the scope of this paper.

At the one-year inflation horizon, market prices have effectively already done the Fed’s job, consistent with the broader conclusion of Section 6: the Greenbook’s forecasting value is weakest precisely where forward guidance would matter most.

7 Limitations of Forecasts

7.1 Unexpected Economic Shocks

The most significant limitation of policymaker forecasts is that the forecasts are extremely vulnerable to exogenous shocks. Macroeconomics forecasting models often

rely on historical data patterns and assume that the future economic conditions will follow gradually. However, unexpected events can rapidly disrupt these patterns. Examples include: the COVID-19 pandemic, which caused an unprecedented collapse in economic activity in 2020, the 2008 global financial crisis, which triggered severe contractions in output and employment, geopolitical conflicts that disrupt energy markets and global supply chains. These events are inherently unpredictable and can dramatically alter key macroeconomic variables such as inflation, unemployment, and GDP growth. As a result, even well-designed forecasts can become inaccurate when unexpected shocks occur.

7.2 Behavioral and Institutional Bias

Although the Federal Reserve relies on formal forecasts, such as those provided in the Tealbook, monetary policy decisions still involve judgment by policymakers. Members of the Federal Open Market Committee may interpret the same economic data and forecasts differently, depending on their economic views, policy preferences, and institutional perspectives. Because of these differences, policymakers may place varying levels of importance on certain indicators or risks when evaluating the economic outlook. As a result, the same set of forecasts and data can lead to different interpretations among policymakers, introducing behavioral and institutional biases into the use of forecasts in monetary policy decisions.

7.3 Data Revisions

Forecast accuracy can also be affected by data revisions. Policymakers base their forecasts on the data available at the time the forecast is made. However, many macroeconomic indicators—especially GDP growth and employment statistics—are frequently revised as more complete information becomes available. For example, initial GDP estimates may later be revised significantly as additional economic data is collected. This means that a forecast that appeared inaccurate based on early data may actually have been reasonable given the information policymakers had at the time. Consequently, evaluating forecast performance must account for the fact that policymakers operate under real-time data constraints.

7.4 Structural Changes in the Economy

Another important limitation of policymaker forecasts is that big shifts in how the economy works can make it harder to trust historical data and economic models. Forecasters usually look at how things have related in the past—like how jobs respond to growth or how prices react to demand. But when the economy changes in a major way, those old relationships might not work anymore.

A good example of this happened during the COVID-19 pandemic, when remote work took off and changed the job market almost overnight. Suddenly, lots of

companies went from office-based work to letting people work from anywhere, or at least having hybrid setups. This opened up hiring in new places and changed how people made decisions about work and where to live. It also affected things like housing demand, commuting patterns, and even how we measure productivity. Since most economic models were built on office-era data, these rapid changes made it much harder for policymakers to lean on old trends when trying to predict jobs, productivity, or wages. Looking ahead, another big shift could come from the fast rise of artificial intelligence (AI) and automation. AI might totally change how things are made, what kinds of jobs are needed, and how quickly industries can grow. For example, AI could make work more efficient in areas like finance, healthcare, and tech, but it could also change what skills employers are looking for. If AI really shake up the workforce or how we measure productivity, the old patterns between things like job numbers and economic growth could change too.

Because big changes like remote work or new technology can completely change how the economy works, forecasts based on old data might not be as accurate. That's why policymakers have to keep updating their models and thinking to keep up with how things are evolving.

8 Conclusion

The Fed's forecasting record is mixed. Unemployment nowcasts are impeccable; GDP growth forecasts at the one-year horizon are difficult to distinguish from a two-parameter autoregression. Trend analysis reveals no meaningful improvement in accuracy across the five-decade sample, where slopes appear negative, they are confounded with aggregate reductions in macroeconomic volatility. The Fed consistently outperforms mechanical benchmarks, confirming that the Fed's institutional resources add value, but that value diminishes sharply as the forecast horizon extends and the target variable becomes more volatile. Returning to the motivating question outlined in Section 1; the Fed's credibility as a forecaster is best supported where forecasting is easiest, and weakest where "forward guidance" would matter most. The fact that these findings were surprising perhaps speaks to the performative goal of the Fed overshadowing the need for technical precision.

Overall, policymaker forecasts provide valuable guidance for understanding the future direction of the economy, but their accuracy is limited by several factors. Unexpected shocks, behavioral biases, data revisions, and structural changes in the economy all make economic forecasts inherently uncertain. As a result, while the forecasts serve as a useful tool for guiding policy decisions, they should be interpreted with caution and evaluated alongside other data provided.

A Figures

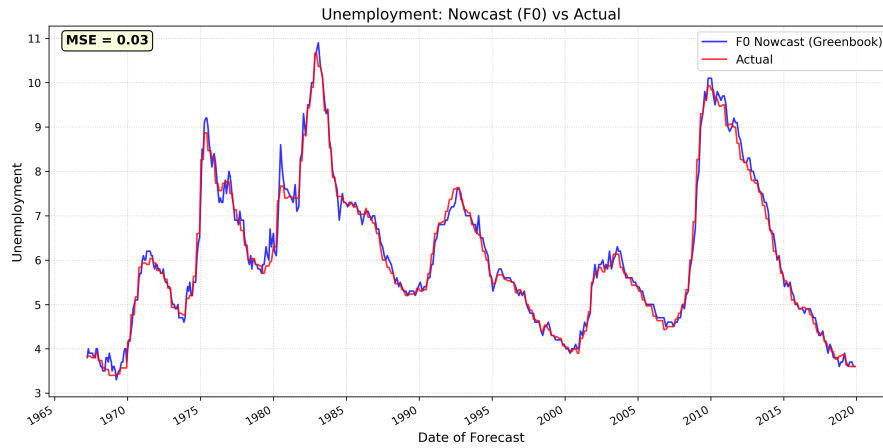


Figure 1

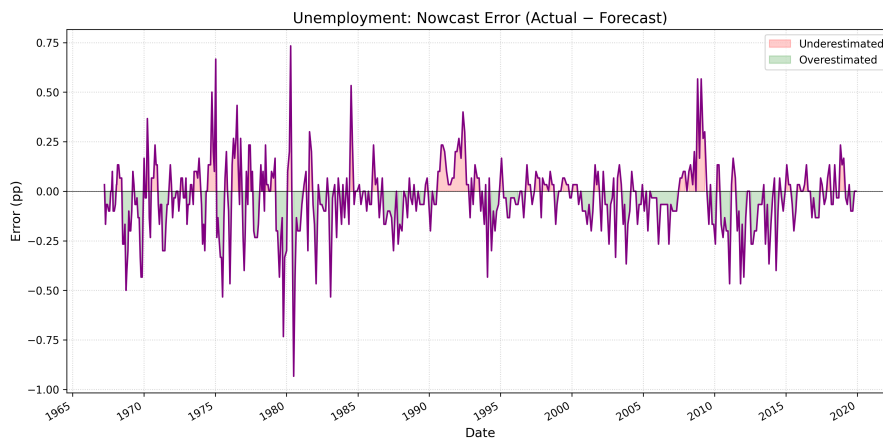


Figure 2

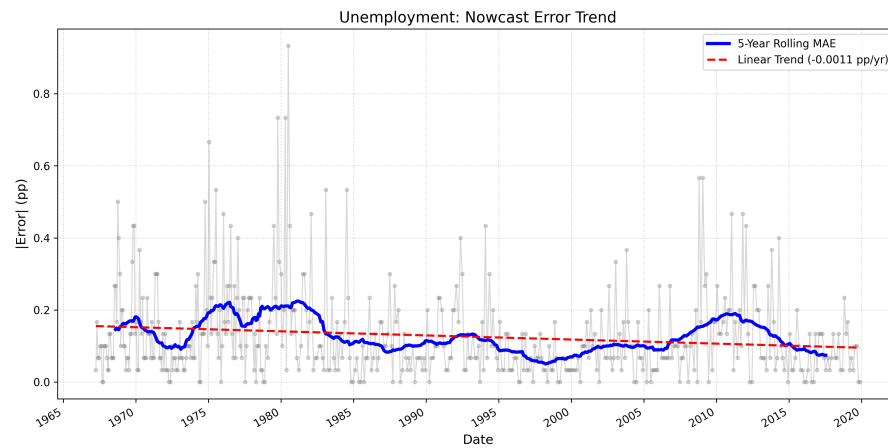


Figure 3

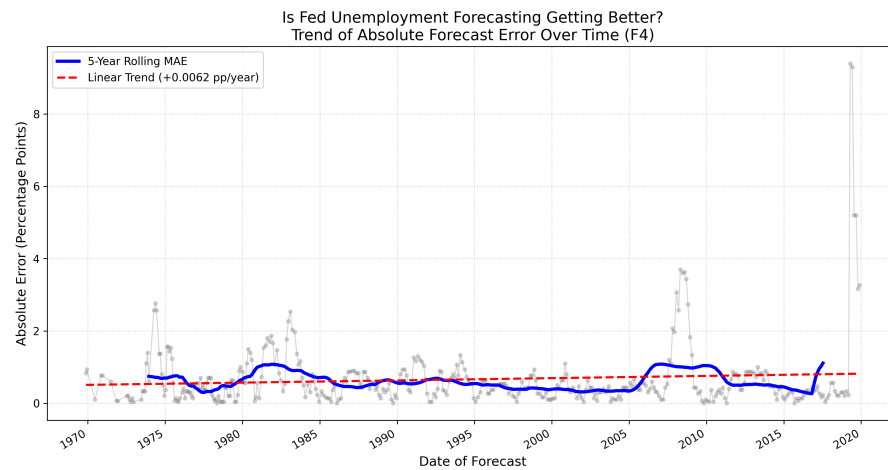


Figure 4

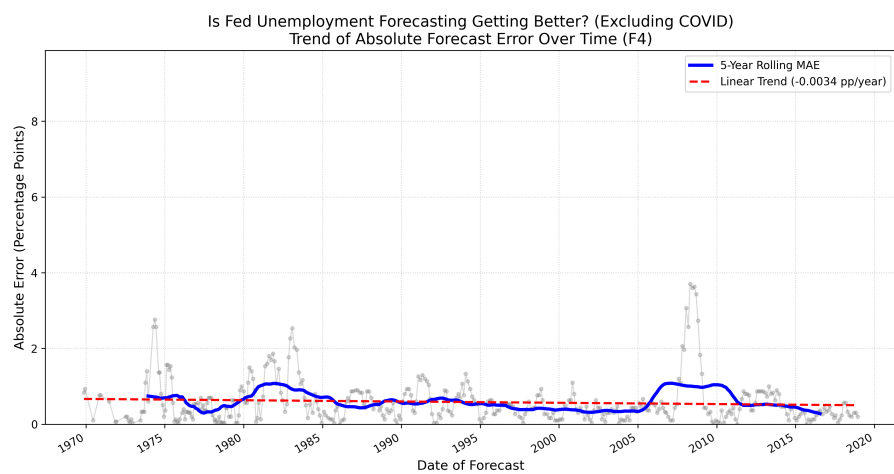


Figure 5

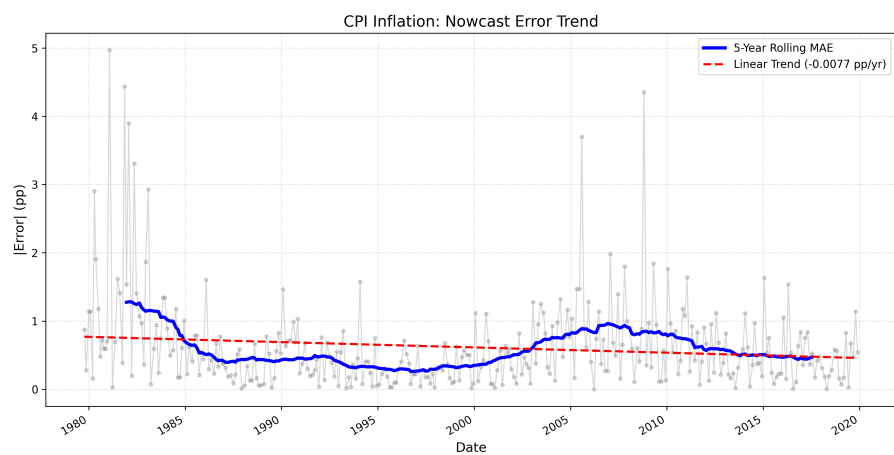


Figure 6

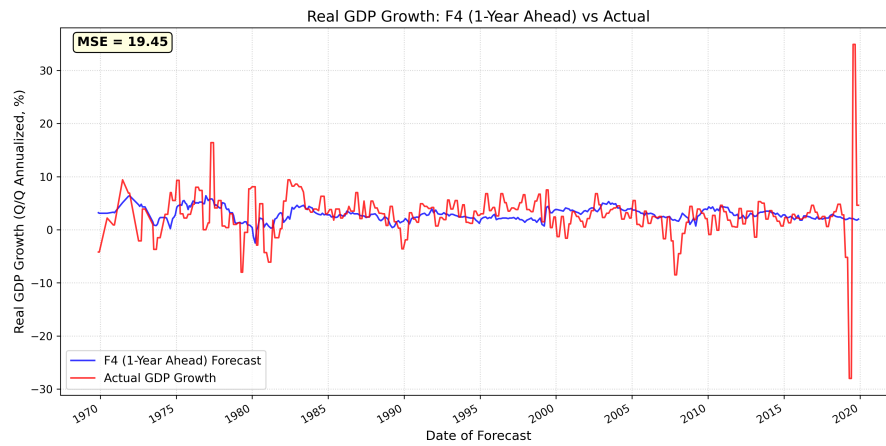


Figure 7

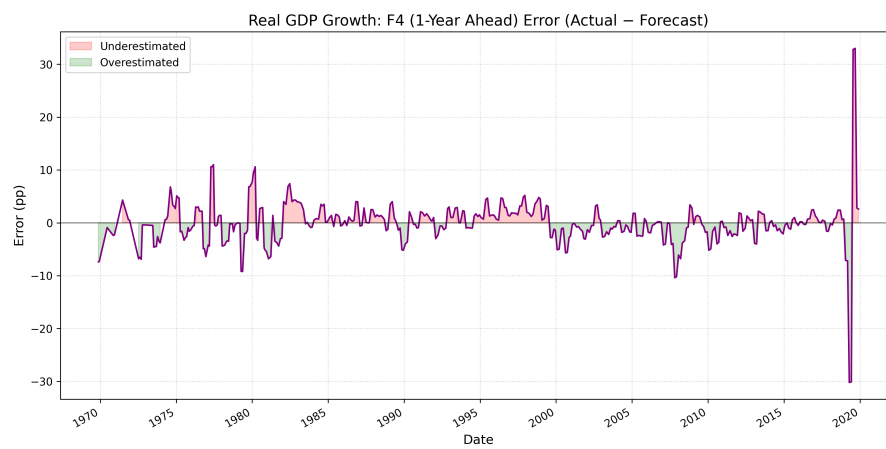


Figure 8

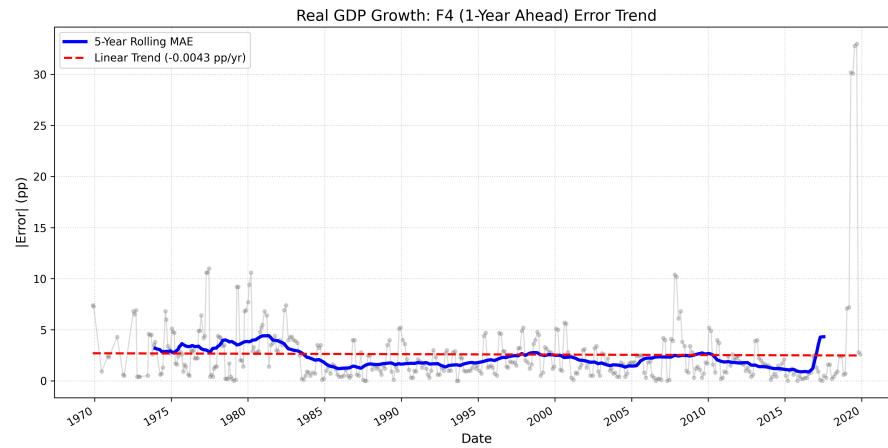


Figure 9

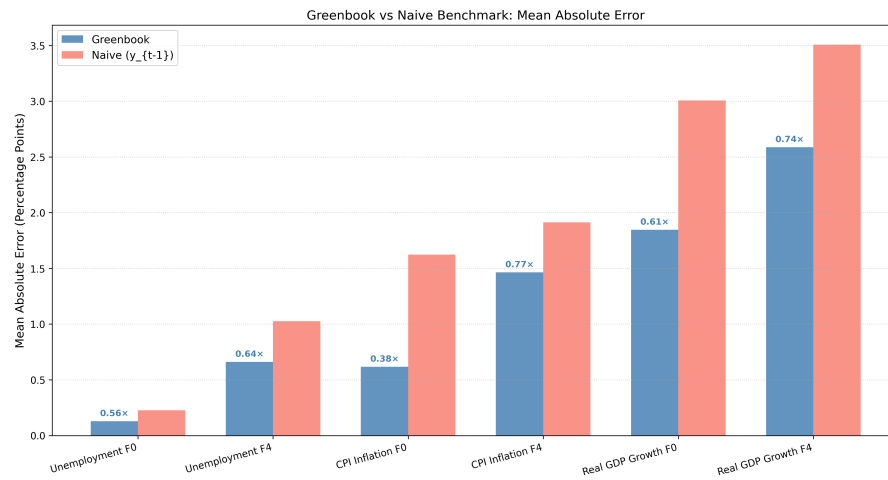


Figure 10

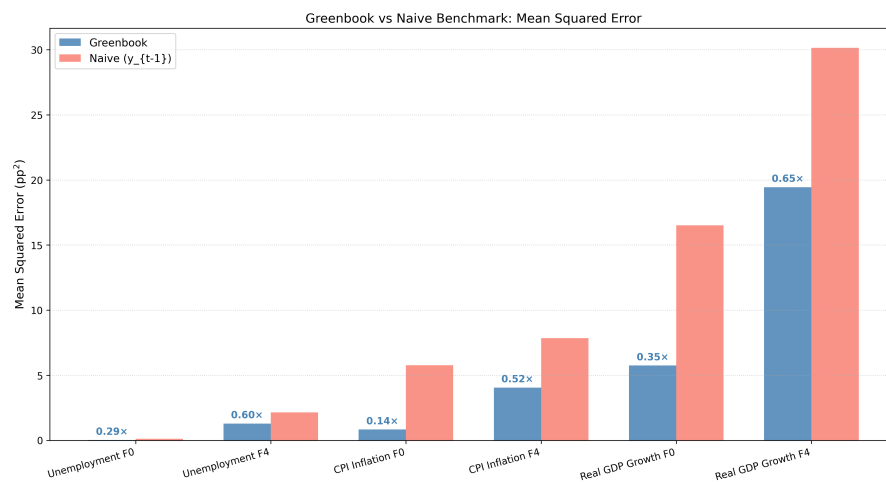


Figure 11

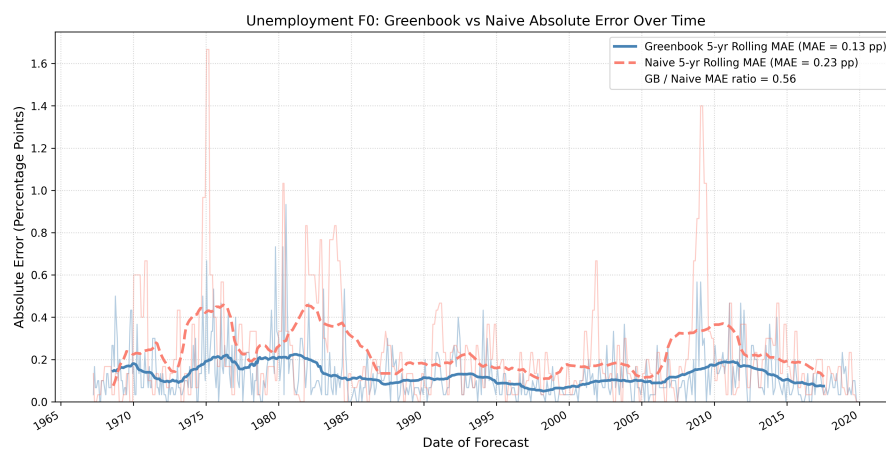


Figure 12

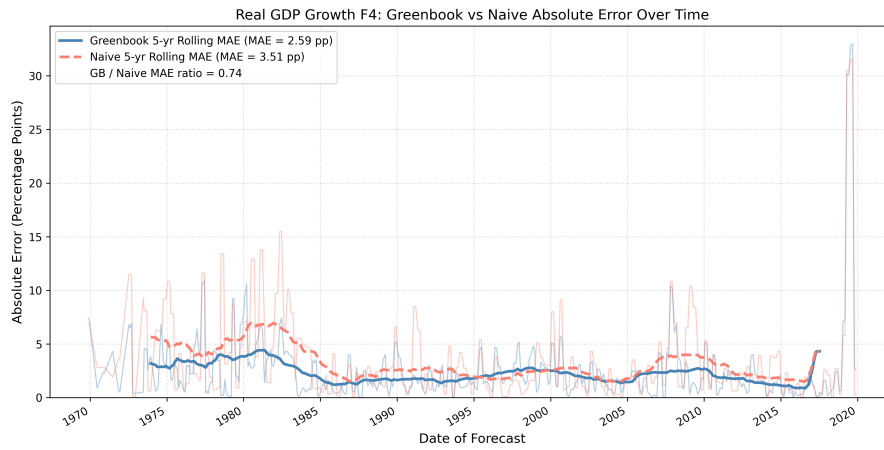


Figure 13

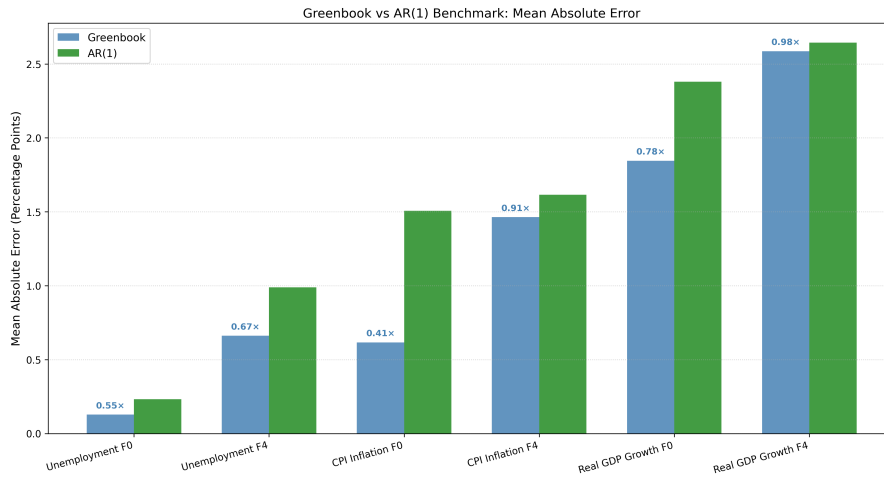


Figure 14

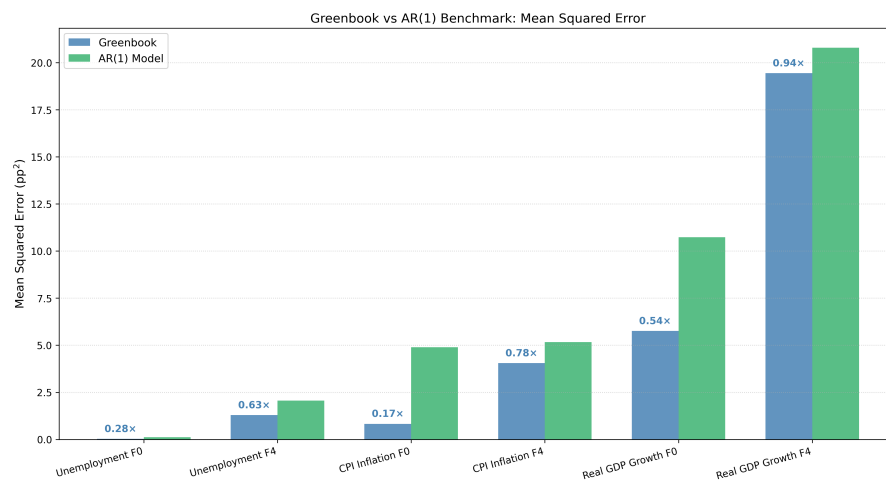


Figure 15

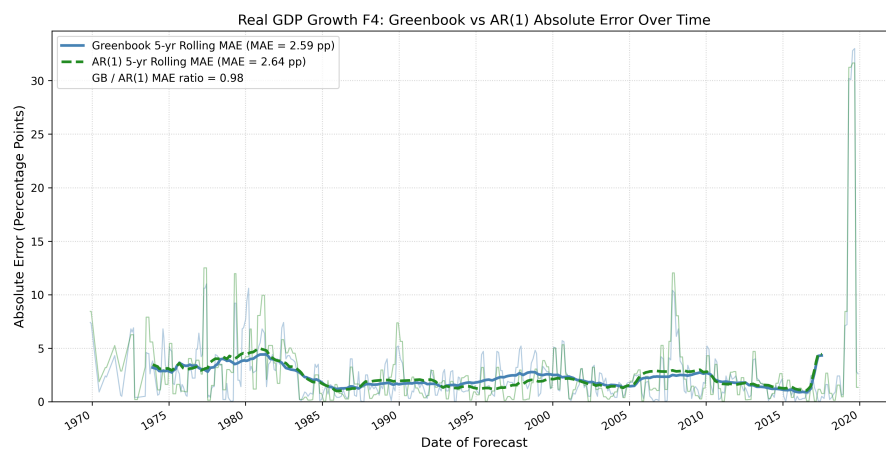


Figure 16

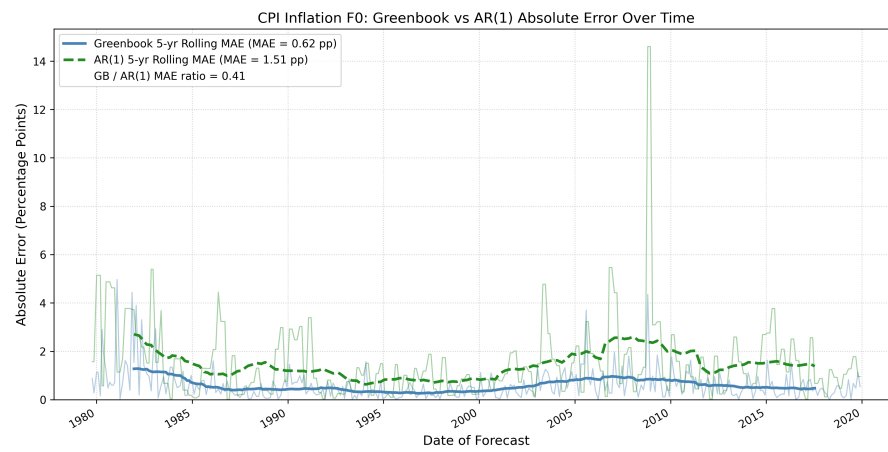


Figure 17